```
> restart;#Pto The torsion particle
> de := 1/V^2 ·H(r)·diff(u(t), t, t) + u(t)·diff(H(r), r, r) = 1/H4(z);
```

$$de := \frac{H(r)\left(\frac{d^2}{dt^2}\,u(t)\right)}{V^2} + u(t)\left(\frac{d^2}{dr^2}\,H(r)\right) = \frac{1}{H4(z)} \tag{1}$$

```
> sol := dsolve(de, u(t));
```

$$sol := u(t) = \frac{e^{_C2\left(\frac{d^2}{dr^2}\,H(r)\right)H4(z)}\cos\left(\frac{\sqrt{H(r)\left(\frac{d^2}{dr^2}\,H(r)\right)}\,V\,(_C1 + t)}{H(r)}\right) + 1}{H4(z)\left(\frac{d^2}{dr^2}\,H(r)\right)} \tag{2}$$

```
> eq1 := r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r) = 0;
```

$$eq1 := r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2\,r - \frac{b^2\,r^2}{R}\right)H3(r) = 0 \tag{3}$$

```
> eq2 := (r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r))/r = 0;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2\,r - \frac{b^2\,r^2}{R}\right)H3(r)}{r} = 0 \tag{4}$$

```
>  # Thus with R:=3.566·10^22 m we can ignore the therm - b^2·r/R for 1/R < r < R;
```

$$\frac{1}{R} < r < R \tag{5}$$

```
> 1/r diff((r·diff H(r), r), r) + k^2· H(r) = 0;
```

$$\frac{2\,diff\left(\frac{d}{dr}\,H(r)\right) + r\,diff\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2\,H(r) = 0 \tag{6}$$

```
> de := 1/r·diff(r·diff(H(r), r), r) + k^2·H(r);
```

$$de := \frac{\frac{d}{dr}\,H(r) + r\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2\,H(r) \tag{7}$$

```
> bc := H(2 R) = 0, D(H)(1/R) = R;
```

$$bc := H(2\,R) = 0,\ D(H)\left(\frac{1}{R}\right) = R \tag{8}$$

```
> dsol := dsolve( {de, bc}, H(r) );
```

$$dsol := H(r) = \frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \tag{9}$$

```
> H(r) := R*BesselY(0, 2*k*R)*BesselJ(0, k*r)/(k*(BesselJ(0, 2*k*R)*BesselY(1, k/R) - BesselY(0, 2*k*R)*BesselJ(1, k/R))) - BesselJ(0, 2*k*R)*R*BesselY(0, k*r)/(k*(BesselJ(0, 2*k*R)*BesselY(1, k/R) - BesselY(0, 2*k*R)*BesselJ(1, k/R)));
```

$$H := r \mapsto \frac{R\cdot\mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot R\cdot\mathrm{BesselY}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \tag{10}$$

```
> u(t) := (exp(_C2*(diff(H(r), r$2))*H4(z))*cos(sqrt((diff(H(r), r$2))*H(r))*V(_C1 + t)/H(r)) + 1)/((diff(H(r), r$2))*H4(z));
```

$$u := t \mapsto \frac{e^{_C2\cdot\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H4(z)}\cdot\cos\left(\frac{\sqrt{\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H(r)}\cdot V\cdot(_C1 + t)}{H(r)}\right) + 1}{\left(\frac{d^2}{dr^2}\,H(r)\right)\cdot H4(z)} \tag{11}$$

```
> B := subs(H(r) = R*BesselY(0, k*R)*BesselJ(0, k*r)/(k*(BesselJ(0, k*R)*BesselY(1, k/R) - BesselY(0, k*R)*BesselJ(1, k/R))) - BesselJ(0, k*R)*R*BesselY(0, k*r)/(k*(BesselJ(0, k*R)*BesselY(1, k/R) - BesselY(0, k*R)*BesselJ(1, k/R))), u(t)/r);
```

$$B := \tag{12}$$

$$\left(e^{_C2 \left(- \frac{R\,\mathrm{BesselY}(0,\,2\,k\,R)\left(\mathrm{BesselJ}(0,\,k\,r) - \frac{\mathrm{BesselJ}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,2\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)} \right.} \right.$$

$$\left. \left. + \frac{\mathrm{BesselJ}(0,\,2\,k\,R)\,R\left(\mathrm{BesselY}(0,\,k\,r) - \frac{\mathrm{BesselY}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,2\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)} \right)$$

$$H4(z)\,\cos\left(\left(\left(\left(\right.\right.\right.\right.$$

$$\left. - \frac{R\,\mathrm{BesselY}(0,\,2\,k\,R)\left(\mathrm{BesselJ}(0,\,k\,r) - \frac{\mathrm{BesselJ}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,2\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)} \right.$$

$$\left. + \frac{\mathrm{BesselJ}(0,\,2\,k\,R)\,R\left(\mathrm{BesselY}(0,\,k\,r) - \frac{\mathrm{BesselY}(1,\,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,\,2\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,2\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)} \right)$$

$$\left(\frac{R\,\mathrm{BesselY}(0,\,k\,R)\,\mathrm{BesselJ}(0,\,k\,r)}{k\left(\mathrm{BesselJ}(0,\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)} \right.$$

$$\left. - \frac{\mathrm{BesselJ}(0,\,k\,R)\,R\,\mathrm{BesselY}(0,\,k\,r)}{k\left(\mathrm{BesselJ}(0,\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)} \right)\Bigg)\Bigg)^{1/2} V(_C1$$

$$+ t)\Bigg) \Bigg/ \left(\frac{R\,\mathrm{BesselY}(0,\,k\,R)\,\mathrm{BesselJ}(0,\,k\,r)}{k\left(\mathrm{BesselJ}(0,\,k\,R)\,\mathrm{BesselY}\left(1,\,\frac{k}{R}\right) - \mathrm{BesselY}(0,\,k\,R)\,\mathrm{BesselJ}\left(1,\,\frac{k}{R}\right)\right)} \right.$$

$$-\frac{\mathrm{BesselJ}(0,kR)\,R\,\mathrm{BesselY}(0,kr)}{k\left(\mathrm{BesselJ}(0,kR)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,kR)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\Bigg)\Bigg)+1\Bigg)$$

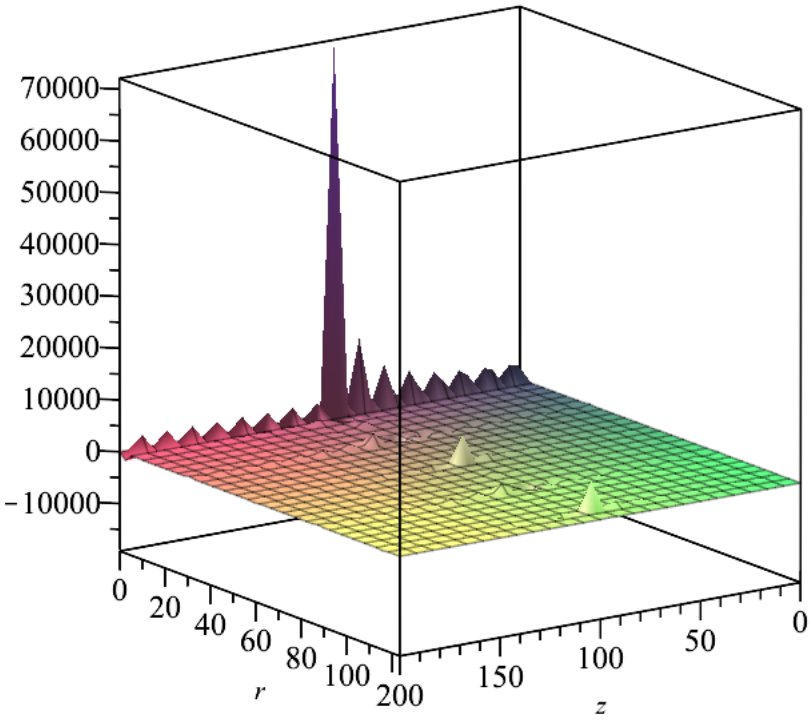
$$\Bigg/\Bigg(\Bigg(\Bigg(-\frac{R\,\mathrm{BesselY}(0,2kR)\left(\mathrm{BesselJ}(0,kr)-\frac{\mathrm{BesselJ}(1,kr)}{kr}\right)k}{\mathrm{BesselJ}(0,2kR)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2kR)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}$$

$$+\frac{\mathrm{BesselJ}(0,2kR)\,R\left(\mathrm{BesselY}(0,kr)-\frac{\mathrm{BesselY}(1,kr)}{kr}\right)k}{\mathrm{BesselJ}(0,2kR)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2kR)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\Bigg)H4(z)\,r\Bigg)$$

> $R := 3.7\cdot 10^3;\ _C1 := 1;\ _C2 := 1;\ k := 1;\ V := 30;\ H4(z) := \cos(z);$

$$R := 3700.0$$
$$_C1 := 1$$
$$_C2 := 1$$
$$k := 1$$
$$V := 30$$
$$H4 := z \mapsto \cos(z) \qquad \textbf{(13)}$$

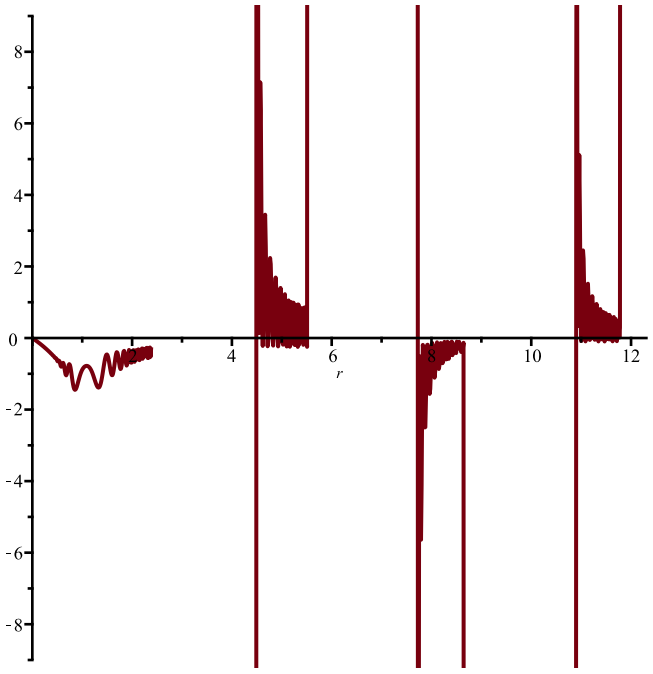
> $plot3d\left(subs(t=0, B), z=0..200, r=1/R..\frac{R}{30}\right);$



> $t := 10; z := 0; plot\left(B, r = 1/R .. \frac{R}{300}, view = -9 .. 9, numpoints = 1000\right);$

$$t := 10$$

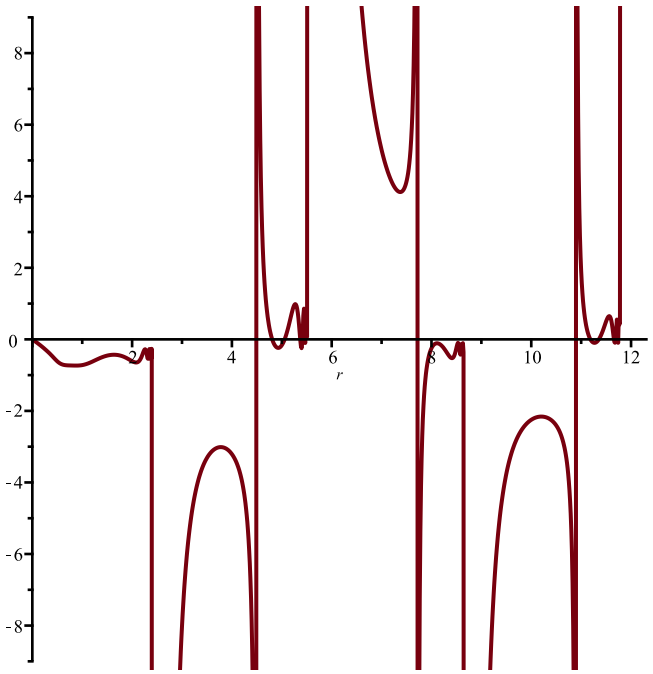
$$z := 0$$



> $t := 0; z = 1; plot\left(B, r = 1/R .. \frac{R}{300}, view = -9 .. 9, numpoints = 1000\right);$

$$t := 0$$

$$0 = 1$$



>